15/4/2026

Recall: Gamma & Beta fns.

$$\Gamma(k) = \int_0^\infty e^{-x} x^{k-1}\,dx$$

$$= 2\int_0^\infty e^{-x^2} x^{2k-1}\,dx$$

Eg: Evaluate $\int_0^\infty e^{-2x^3}\,dx$

put $2x^3 = t$

$\Rightarrow x^3 = \frac{t}{2}$

$\Rightarrow x = \left(\frac{t}{2}\right)^{1/3}$

$\Rightarrow dx = \frac{1}{3\,2^{1/3}} \cdot t^{-2/3}\,dt$

x	0	∞
$t = 2x^3$	0	∞

$$\Rightarrow I = \int_0^\infty e^{-t} \cdot \frac{1}{3} \cdot \frac{1}{2^{1/3}}\, t^{-2/3}\,dt$$

$$= \frac{1}{2^{1/3} \cdot 3} \underbrace{\int_0^\infty e^{-t}\, t^{-2/3}\,dt}_{\Gamma\left(\frac{1}{3}\right)} \quad k = -\frac{2}{3}+1$$

$$\Rightarrow I = \frac{1}{3 \cdot 2^{1/3}}\,\Gamma(1/3)$$

2. Evaluate $\int_0^\infty \frac{x^4}{4^x}\,dx$

Sol^n:

put $4^x = e^t$

$$\left[\log_a b = \frac{\log_e b}{\log_e a}\right]$$

$\Rightarrow x \log 4 = t \log e$

$\Rightarrow x \log 4 = t$

$\Rightarrow x = \frac{t}{\log 4}$

$\Rightarrow dx = \frac{dt}{\log 4}$

$\longleftrightarrow e^{-x}$

x	0	∞
$t = x\log 4$	0	∞

$$\therefore I = \int_0^\infty \left(\frac{t}{\log 4}\right)^4 \cdot \frac{1}{e^t}\,\frac{dt}{\log 4}$$

$$= \int_0^\infty t^4 \cdot e^{-t} \cdot \frac{1}{(\log 4)^5}\,dt$$

$$= \frac{1}{(\log 4)^5}\int_0^\infty e^{-t}\,t^4\,dt = \frac{1}{(\log 4)^5}\,\Gamma(5) = 4!$$

Beta function

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

is called the Beta function (integral) with parameters m & n.

Properties of Beta fn.

1. $\beta(m,n) = \beta(n,m)$

Proof: $\left[\text{LHS} = \beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx\right.$

& $\text{RHS} = \beta(n,m) = \int_0^1 x^{n-1}(1-x)^{m-1}\,dx$

$$\Gamma(k) = \int_0^\infty e^{-x}\,x^{k-1}\,dx$$

put $1-x = t$

$\Rightarrow x = 1-t$

$\Rightarrow dx = -dt$

x	0	1
$t = 1-x$	1	0

$$\therefore \beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

$$= \int_1^0 (1-t)^{m-1}\,t^{n-1}(-dt)$$

$$= \int_0^1 t^{n-1}(1-t)^{m-1}\,dt \qquad \left[\because \int_a^b -dt = \int_b^a dt\right]$$

$$= \beta(n,m)$$

2. $\beta(m,n) = \dfrac{\Gamma(m)\cdot\Gamma(n)}{\Gamma(m+n)}$

<u>Eg:</u> Find

① $\beta(2,3) = \dfrac{\Gamma(2)\cdot\Gamma(3)}{\Gamma(2+3)}$

$= \dfrac{1!\ 2!}{4!}$

$\Rightarrow \beta(2,3) = \dfrac{1}{12}$

2. Find $\beta\left(\frac{3}{2},\frac{1}{2}\right)$

$\beta\left(\frac{3}{2},\frac{1}{2}\right) = \dfrac{\Gamma\left(\frac{3}{2}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}+\frac{1}{2}\right)}$

$= \dfrac{\frac{1}{2}\cdot\Gamma\left(\frac{1}{2}\right)\cdot\Gamma(1/2)}{\Gamma(2)}$

$\left[\because \Gamma(k) = (k-1)\Gamma(k-1)\right.$

$\Rightarrow \beta\left(\frac{3}{2},\frac{1}{2}\right) = \dfrac{\frac{1}{2}\sqrt{\pi}\sqrt{\pi}}{1!}$

$= \dfrac{\pi}{2}$ //

3. Find $\beta\left(\frac{5}{2},\frac{7}{2}\right)$

$\beta\left(\frac{5}{2},\frac{7}{2}\right) = \dfrac{\Gamma\left(\frac{5}{2}\right)\cdot\Gamma\left(\frac{7}{2}\right)}{\Gamma(6)}$

$= \dfrac{\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2)}{5!}$

$= \dfrac{45\pi}{32\times 5!}$

$= \dfrac{3\pi}{256}$ //

4. Evaluate $\int_0^a x^m (a-x)^n dx$

<u>Sol^n</u>: We have

$$a - x = a\left(1 - \frac{x}{a}\right)$$

put $\frac{x}{a} = t$

$\Rightarrow x = at$

$\Rightarrow dx = a\,dt$

x	0	a
$t = \frac{x}{a}$	0	1

$$\therefore \int_0^a x^m (a-x)^n dx = \int_0^1 (at)^m (a - at)^n \cdot a\,dt$$

$$\Rightarrow I = \int_0^1 a^m t^m \left[a(1-t)\right]^n \cdot a\,dt$$

$$= a^{m+1} \int_0^1 t^m a^n (1-t)^n \cdot dt$$

$$= a^{m+n+1} \int_0^1 t^m (1-t)^n dt$$

$$\downarrow$$

$$= a^{m+n+1} \beta(m+1, n+1)$$

5. Evaluate $\int_0^1 x^m (1-x^n)^p \, dx$

<u>Soln</u>: put $x^n = t$

$\Rightarrow x = t^{1/n}$

$\Rightarrow dx = \frac{1}{n} t^{\frac{1}{n}-1} \, dt$

x	0	1
$t = x^n$	0	1

$\therefore \int_0^1 x^m (1-x^n)^p \, dx$

$= \int_0^1 \left(t^{1/n}\right)^m (1-t)^p \cdot \frac{1}{n} t^{\frac{1}{n}-1} \, dt$

$\Rightarrow I = \frac{1}{n} \int_0^1 t^{\frac{m}{n}} (1-t)^p \cdot t^{\frac{1}{n}-1} \, dt$

$= \frac{1}{n} \int_0^1 t^{\frac{m+1}{n}-1} (1-t)^p \, dt$

$= \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right)$

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